

Arithmetic Sequences

Algebra Worksheet · Grade 8–10

Name: _____

Date: _____

Learning Objectives

- Identify whether a sequence is arithmetic by finding common differences
- Use the n th-term formula to find specific terms of an arithmetic sequence
- Derive the formula for the n th term given the first term and common difference

Problems

1. Find the common difference of the arithmetic sequence below.

3, 7, 11, 15, 19

2. Determine whether the following sequence is arithmetic. Explain your reasoning.

2, 5, 9, 14, 20

3. Determine whether the following sequence is arithmetic. If yes, state the common difference.

20, 15, 10, 5, 0

4. Write the next two terms of the arithmetic sequence below.

5. Find the n th-term formula for an arithmetic sequence with first term 5 and common difference 3.

$a_1 = 5, \quad d = 3$

6. Find the n th-term formula for an arithmetic sequence with first term 20 and common difference negative 4.

$a_1 = 20, \quad d = -4$



7. Find the 15th term of the arithmetic sequence below.

5, 8, 11, 14, ...

8. Find the 30th term of the arithmetic sequence below.

10, 7, 4, 1, ...

9. The 5th term of an arithmetic sequence is 27 and the common difference is 4. Find the first term and then write the nth-term formula.

$$a_5 = 27, \quad d = 4$$

10. The 4th term of an arithmetic sequence is 3 and the 10th term is 33. Find the common difference, the first term, and the 20th term.

$$a_4 = 3, \quad a_{10} = 33$$



Arithmetic Sequences — Answer Key

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Answer Key

1. Answer: $d = 4$

- Subtract consecutive terms: $7 - 3 = 4$
 - Check: $11 - 7 = 4$, $15 - 11 = 4$, $19 - 15 = 4$
 - The common difference is 4
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2. Answer: Not arithmetic

- Find differences: $5 - 2 = 3$, $9 - 5 = 4$, $14 - 9 = 5$
 - The differences are not the same
 - Since there is no common difference, this is NOT an arithmetic sequence
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3. Answer: Arithmetic; $d = -5$

- Find differences: $15 - 20 = -5$, $10 - 15 = -5$, $5 - 10 = -5$
 - All differences are equal
 - This is arithmetic with a common difference of -5
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4. Answer: 26 and 31

- Find the common difference: $11 - 6 = 5$
 - Add 5 to 21: $21 + 5 = 26$
 - Add 5 to 26: $26 + 5 = 31$
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5. Answer: $a_n = 3n + 2$

- Use the formula: $a_n = a_1 + (n - 1)d$
 - Substitute: $a_n = 5 + (n - 1)(3)$
 - Distribute: $a_n = 5 + 3n - 3$
 - Simplify: $a_n = 3n + 2$
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6. Answer: $a_n = -4n + 24$

- Use the formula: $a_n = a_1 + (n - 1)d$
 - Substitute: $a_n = 20 + (n - 1)(-4)$
 - Distribute: $a_n = 20 - 4n + 4$
 - Simplify: $a_n = -4n + 24$
-

7. Answer: 47

- Identify: $a_1 = 5$, $d = 3$, $n = 15$
 - Use the formula: $a_n = a_1 + (n - 1)d$
 - Substitute: $a_{15} = 5 + (15 - 1)(3)$
 - Simplify: $a_{15} = 5 + 42 = 47$
-

8. Answer: -77





- Identify: $a_1 = 10$, $d = -3$, $n = 30$
 - Use the formula: $a_n = a_1 + (n - 1)d$
 - Substitute: $a_{30} = 10 + (30 - 1)(-3)$
 - Simplify: $a_{30} = 10 + (-87) = -77$
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9. Answer: $a_1 = 11$; $a_n = 4n + 7$

- Use: $a_5 = a_1 + (5 - 1)(4)$
 - Solve: $27 = a_1 + 16$, so $a_1 = 11$
 - Write formula: $a_n = 11 + (n - 1)(4)$
 - Simplify: $a_n = 11 + 4n - 4 = 4n + 7$
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10. Answer: $d = 5$, $a_1 = -12$, $a_{20} = 83$

- Set up two equations: $a_1 + 3d = 3$ and $a_1 + 9d = 33$
 - Subtract the first from the second: $6d = 30$, so $d = 5$
 - Substitute back: $a_1 + 3(5) = 3$, so $a_1 = -12$
 - Find the 20th term: $a_{20} = -12 + (20 - 1)(5) = -12 + 95 = 83$
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